

## 9.5 The coefficient of determination and its limits

The coefficient of determination is the most widely reported summary of a regression, and probably the most widely misread. It is a ratio of two rows of the corrected table,

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}},$$

defined when  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  and  $\mathbf{y}$  is not constant. [Section 6.7](#) showed that it is a squared cosine (Theorem 6.7.2) that cannot decrease as the model space grows. This section adds  $R^2$  as a maximal correlation, its null distribution, the adjusted and partial versions, and what it does not measure.

### 9.5.1 A maximal correlation

The square root  $R = \sqrt{R^2}$  is called the **multiple correlation coefficient**. The name comes from the following property, the sample counterpart of the population result Proposition 2.5.3.

#### Theorem 9.5.1. $R$ is the largest correlation with a linear combination

Suppose  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  and  $\mathbf{y}$  is not constant. For every  $\mathbf{b}$  such that  $\mathbf{Xb}$  is not constant,

$$\text{corr}(\mathbf{y}, \mathbf{Xb})^2 \leq R^2.$$

If  $R^2 > 0$ , equality holds iff  $\mathbf{Xb} = a\hat{\mathbf{Y}} + c\mathbf{1}$  for some  $a \neq 0$  and some  $c$ , and the correlation  $\text{corr}(\mathbf{y}, \mathbf{Xb})$  is maximized, with value  $R$ , by the fitted values themselves. If  $R^2 = 0$ , every such correlation is zero.

*Proof.* Let  $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$  and  $\mathcal{V} = \mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$ , whose projection is  $\mathbf{M} - \mathbf{P}_1$  (Theorem 6.5.1), and let  $\mathbf{v} = \mathbf{Xb} - \bar{\mathbf{Xb}}\mathbf{1}$ . Since  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ ,  $\mathbf{v} \in \mathcal{V}$ , so  $\mathbf{u}^\top \mathbf{v} = \mathbf{u}^\top (\mathbf{M} - \mathbf{P}_1)\mathbf{v} = ((\mathbf{M} - \mathbf{P}_1)\mathbf{u})^\top \mathbf{v}$ . The sample correlation is  $\mathbf{u}^\top \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$ , and by Cauchy–Schwarz (Proposition 1.1.1)

$$\text{corr}(\mathbf{y}, \mathbf{Xb})^2 = \frac{((\mathbf{M} - \mathbf{P}_1)\mathbf{u})^\top \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \leq \frac{\|(\mathbf{M} - \mathbf{P}_1)\mathbf{u}\|^2}{\|\mathbf{u}\|^2} = \frac{\text{SSR}}{\text{SST}},$$

with equality iff  $\mathbf{v}$  and  $(\mathbf{M} - \mathbf{P}_1)\mathbf{u} = \hat{\mathbf{Y}} - \bar{y}\mathbf{1}$  are linearly dependent. If  $R^2 > 0$  the second vector is nonzero, and since  $\mathbf{v} \neq \mathbf{0}$  dependence means  $\mathbf{v} = a(\hat{\mathbf{Y}} - \bar{y}\mathbf{1})$  with  $a \neq 0$ , which is the stated condition; for  $a > 0$  the correlation itself is  $+R$ . If  $R^2 = 0$ , then  $(\mathbf{M} - \mathbf{P}_1)\mathbf{u} = \mathbf{0}$  and  $\mathbf{u}^\top \mathbf{v} = 0$ . ■

Among all linear combinations of the regressors, the fitted values are the one most correlated with the response. When the regressors are pairwise uncorrelated,  $R^2$  is the sum of the squared simple correlations (Exercise (A3)). In general it can be smaller or larger than that sum ([Exercise \(C2\)](#)).

## 9.5.2 $R^2$ when nothing is going on

Since  $R^2$  never decreases as regressors are added, a model with many regressors has a sizeable  $R^2$  even if none of them matters. The theory of [Chapter 4](#) says how sizeable.

### Theorem 9.5.2. The distribution of $R^2$

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$  with  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ ,  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  and  $1 < r = \text{rank}(\mathbf{X}) < n$ . Let  $F = [\text{SSR}/(r-1)]/[\text{SSE}/(n-r)]$  and  $\gamma = \|(\mathbf{M} - \mathbf{P}_1)\boldsymbol{\mu}\|^2/\sigma^2$ . Then:

- (a)  $R^2 = \frac{(r-1)F}{(r-1)F + n-r}$ , and  $F \sim F(r-1, n-r, \gamma)$ ;
- (b) if  $\boldsymbol{\mu} \in \text{span}(\mathbf{1})$ , so that no regressor matters, then  $R^2 \sim \text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$  and  $E(R^2) = \frac{r-1}{n-1}$ .

*Proof.*

- (a)  $R^2/(1-R^2) = \text{SSR}/\text{SSE} = (r-1)F/(n-r)$ , and solving for  $R^2$  gives the formula. The projections  $\mathbf{M} - \mathbf{P}_1$  and  $\mathbf{I} - \mathbf{M}$  are orthogonal with ranks  $r-1$  and  $n-r$ , and  $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \mathbf{0}$ , so by Theorem 4.4.4  $\text{SSR}/\sigma^2 \sim \chi^2(r-1, \gamma)$  and  $\text{SSE}/\sigma^2 \sim \chi^2(n-r)$ , independently. The ratio  $F$  is therefore  $F(r-1, n-r, \gamma)$  (Definition 4.1.2).
- (b) Now  $\gamma = 0$ , and  $R^2 = U/(U+V)$  with  $U = \text{SSR}/\sigma^2 \sim \chi^2(r-1)$  and  $V = \text{SSE}/\sigma^2 \sim \chi^2(n-r)$  independent. By Exercise (B4) this is  $\text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$ , whose mean is  $\frac{r-1}{2} / \frac{n-1}{2} = (r-1)/(n-1)$ . ■

Part (b) answers the exercise Exercise (C1) of [Chapter 6](#). With  $k = r-1$  regressors of pure noise,  $R^2$  is on average  $k/(n-1)$ . A model with 10 useless regressors fitted to 30 observations has expected  $R^2$  about one third. [Figure 9.5.1](#) compares the Beta densities with 100,000 simulated data sets with  $n = 30$  and  $k = 1, 5, 10$  regressors drawn at random. The simulated means are 0.0345, 0.1725 and 0.3445, against the exact 0.0345, 0.1724 and 0.3448. The upper 5% points of the three null distributions are 0.130, 0.353 and 0.556, so with ten regressors an  $R^2$  of one half is unremarkable. Part (a) shows that  $R^2$  and  $F$  carry the same information: testing  $R^2$  against its null distribution is the overall  $F$  test of the regression.

```
import numpy as np
rng = np.random.default_rng(20260919)
n, reps = 30, 100_000
results = {}
for k in (1, 5, 10):                                     # number of regressors
    besides 1
    X = np.column_stack([np.ones(n), rng.normal(size=(n, k))])
    r = k + 1
    Q, _ = np.linalg.qr(X)
    Y = 3.0 + rng.normal(size=(n, reps))                 # no regressor matters
    Z = Q.T @ Y
    ssr = np.sum(Z[1:] ** 2, axis=0)
    sst = np.sum(Y ** 2, axis=0) - Z[0] ** 2
    R2 = ssr / sst
    R2adj = 1 - (1 - R2) * (n - 1) / (n - r)
```

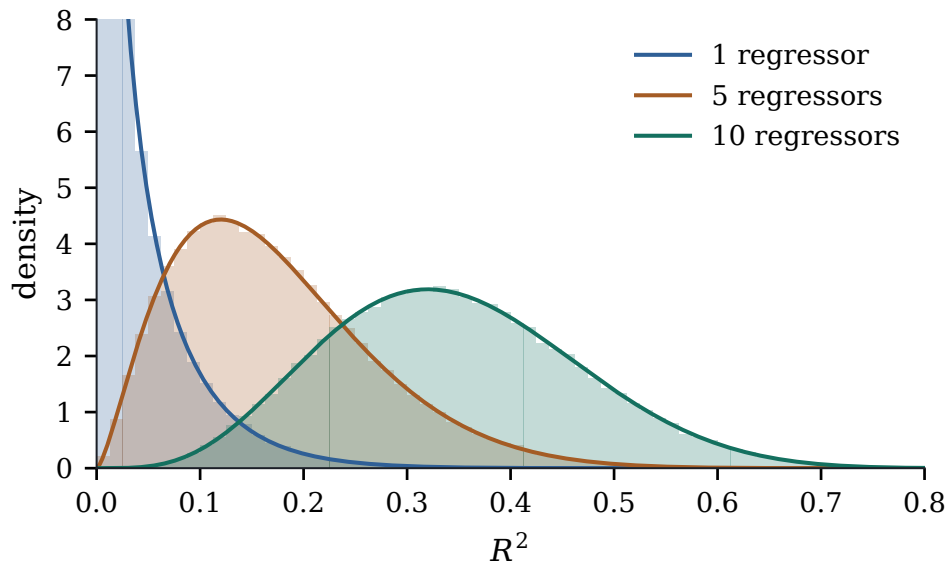


Figure 9.5.1: Null distribution of  $R^2$  for  $n = 30$  observations and 1, 5 or 10 regressors unrelated to the response. Histograms: 100,000 simulated data sets each. Curves: the Beta densities of Theorem 9.5.2.

```
print(f"k = {k:2d}: mean R^2 = {R2.mean():.4f} (theory {k / (n - 1):.4f}),"
      f" mean adjusted R^2 = {R2adj.mean():+.4f}")
results[k] = R2
```

[Run this code online](#)

### 9.5.3 Adjusted $R^2$

The bias of  $R^2$  under the null suggests a correction. Write  $s^2 = \text{SSE}/(n-r)$  and  $s_y^2 = \text{SST}/(n-1)$  for the unbiased estimators of the error variance in the fitted model and in the intercept-only model. Replacing sums of squares by these mean squares defines the **adjusted coefficient of determination**

$$\bar{R}^2 = 1 - \frac{s^2}{s_y^2} = 1 - (1 - R^2) \frac{n-1}{n-r}. \quad (9.5.1)$$

#### Proposition 9.5.3. Properties of the adjusted $R^2$

Let  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ ,  $1 \leq r = \text{rank}(\mathbf{X}) < n$ , and let  $F$  be the overall  $F$  ratio of Theorem 9.5.2.

- (a)  $\bar{R}^2 \leq R^2$ , with equality iff  $r = 1$  or  $R^2 = 1$ .
- (b) For  $r > 1$ ,  $\bar{R}^2 < 0$  iff  $F < 1$ .
- (c) If a model with rank  $r_0$  is enlarged to one of rank  $r = r_0 + q$  containing it, then  $\bar{R}^2$  increases iff the  $F$  ratio of the added terms,  $[\text{SS}(\text{added} \mid \text{old})/q]/s^2$ , exceeds 1. For a single added column this means  $|t| > 1$ .
- (d) Among models for the same response,  $\bar{R}^2$  is largest for the model with the smallest residual mean square  $s^2$ .

(e) Under the normal model with  $\mu \in \text{span}(\mathbf{1})$ ,  $E(\bar{R}^2) = 0$ .

*Proof.*

- (a)  $1 - \bar{R}^2 = (1 - R^2)(n - 1)/(n - r)$ , and  $(n - 1)/(n - r) \geq 1$  with equality iff  $r = 1$ , so  $1 - \bar{R}^2 \geq 1 - R^2$ , with equality iff  $r = 1$  or  $1 - R^2 = 0$ .
- (b)  $\bar{R}^2 < 0$  iff  $s^2 > s_y^2$ , that is,  $\text{SSE}(n - 1) > (\text{SSR} + \text{SSE})(n - r)$ , that is,  $\text{SSE}(r - 1) > \text{SSR}(n - r)$ , which for  $r > 1$  is  $F < 1$ .
- (c)  $s_y^2$  does not depend on the model, so  $\bar{R}^2$  increases iff  $s^2$  decreases. Write  $\text{SSE}_0 = \text{SSE} + D$  with  $D$  the extra sum of squares. Then  $\text{SSE}/(n - r) < (\text{SSE} + D)/(n - r_0)$  iff  $\text{SSE}(n - r_0) < (\text{SSE} + D)(n - r)$ , iff  $q \text{SSE} < D(n - r)$ , iff  $(D/q)/s^2 > 1$ . For one column the ratio is  $t^2$  (Corollary 9.2.3).
- (d) is immediate from Eq. 9.5.1.
- (e) By Theorem 9.5.2(b),  $E(1 - R^2) = 1 - (r - 1)/(n - 1) = (n - r)/(n - 1)$ , so  $E(1 - \bar{R}^2) = 1$ .

■

Part (e) is the point of the adjustment. In the simulation the averages of  $\bar{R}^2$  are 0.0000, 0.0001 and  $-0.0005$ , but  $\bar{R}^2$  is negative in a proportion 0.674 of the data sets with one regressor: it is not a proportion of anything. Part (c) shows what selection by  $\bar{R}^2$  amounts to: a regressor is admitted when  $|t| > 1$ , far below the threshold near two of a 5% test, so the chosen models tend to be too large (Proposition 29.2.3).

#### Example. Adjusted $R^2$ for the state data

For all seven models built from poverty, single parenthood and urbanization (each with an intercept):

| Regressors      | $R^2$  | $\bar{R}^2$ |
|-----------------|--------|-------------|
| poverty         | 0.3599 | 0.3466      |
| single          | 0.5719 | 0.5629      |
| urban           | 0.0221 | 0.0017      |
| poverty, single | 0.6410 | 0.6257      |
| poverty, urban  | 0.4626 | 0.4398      |
| single, urban   | 0.5844 | 0.5668      |
| all three       | 0.6419 | 0.6186      |

$R^2$  is largest for the full model, as it must be.  $\bar{R}^2$  prefers poverty and single parenthood without urbanization. Adding urbanization to that model raises  $R^2$  slightly but lowers  $\bar{R}^2$ , because its  $t$  statistic in the full model is only 0.349, so its  $F$  ratio is  $0.121 < 1$ .

### 9.5.4 Partial $R^2$

Applied to what a reduced model leaves unexplained, the idea of  $R^2$  gives a measure for a term.

**Definition 9.5.1. Partial coefficient of determination**

Let  $\mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X})$  with  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$ . The **partial coefficient of determination** of  $\mathbf{X}$  over  $\mathbf{X}_1$  is

$$R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{\text{SSE}(\mathbf{X}_1) - \text{SSE}(\mathbf{X})}{\text{SSE}(\mathbf{X}_1)} = \frac{\text{SS}(\mathbf{X} | \mathbf{X}_1)}{\text{SSE}(\mathbf{X}_1)},$$

the proportion of the residual variation of the reduced model that the larger model explains.

**Theorem 9.5.4. Partial  $R^2$ ,  $F$  and  $t$** 

With the notation of [Definition 9.5.1](#), let  $r_1 = \text{rank}(\mathbf{X}_1)$ ,  $r = \text{rank}(\mathbf{X})$ ,  $q = r - r_1 > 0$ ,  $s^2 = \text{SSE}(\mathbf{X})/(n - r)$ , and let  $F = [\text{SS}(\mathbf{X} | \mathbf{X}_1)/q]/s^2$ . Then:

- (a)  $1 - R^2 = (1 - R_1^2)(1 - R_{\mathbf{X}|\mathbf{X}_1}^2)$ , where  $R^2$  and  $R_1^2$  belong to the two models;
- (b)  $R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{qF}{qF + n - r}$ ;
- (c) if  $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}_j]$  adds one column, then  $R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{t_j^2}{t_j^2 + n - r}$ , and it equals the squared sample correlation between the residuals  $(\mathbf{I} - \mathbf{M}_1)\mathbf{y}$  and  $(\mathbf{I} - \mathbf{M}_1)\mathbf{x}_j$ ;
- (d) for a chain  $\mathcal{C}(\mathbf{1}) = \mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_1) \subseteq \cdots \subseteq \mathcal{C}(\mathbf{X}_k)$ ,  $1 - R_k^2 = \prod_{j=1}^k (1 - R_{\mathbf{X}_j|\mathbf{X}_{j-1}}^2)$ .

*Proof.*

- (a)  $1 - R^2 = \text{SSE}(\mathbf{X})/\text{SST} = (\text{SSE}(\mathbf{X}_1)/\text{SST})(\text{SSE}(\mathbf{X})/\text{SSE}(\mathbf{X}_1))$ , and the two factors are  $1 - R_1^2$  and  $1 - R_{\mathbf{X}|\mathbf{X}_1}^2$ .
- (b) Write  $D = \text{SS}(\mathbf{X} | \mathbf{X}_1) = qFs^2$  and  $\text{SSE}(\mathbf{X}) = (n - r)s^2$ . Then  $R_{\mathbf{X}|\mathbf{X}_1}^2 = D/(D + \text{SSE}(\mathbf{X})) = qF/(qF + n - r)$ .
- (c) With  $q = 1$ ,  $F = t_j^2$  by Corollary 9.2.3, and (b) gives the first formula. For the correlation, let  $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_1)\mathbf{y}$  and  $\tilde{\mathbf{x}} = (\mathbf{I} - \mathbf{M}_1)\mathbf{x}_j$ . Then  $\text{SSE}(\mathbf{X}_1) = \|\tilde{\mathbf{y}}\|^2$ , and by Lemma 9.2.1  $\text{SS}(\mathbf{X} | \mathbf{X}_1) = (\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}})^2 / \|\tilde{\mathbf{x}}\|^2$ . Their ratio is the squared cosine of the angle between  $\tilde{\mathbf{y}}$  and  $\tilde{\mathbf{x}}$ . Both have mean zero because  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$ , so the cosine is their sample correlation (Exercise (B3)).
- (d) Apply (a) repeatedly along the chain, with  $R_0^2 = 0$  for the intercept-only model.

■

Part (c) identifies the partial  $R^2$  of one regressor with a squared **sample partial correlation**, the sample counterpart of Definition 3.6.1; [Chapter 14](#) studies its sampling distribution. Within one model, ranking regressors by  $|t|$  and by partial  $R^2$  is the same thing.

**Example. Partial  $R^2$  in the state data**

In the three-regressor model, with  $n - r = 46$ :

| Regressor | $t$   | partial $R^2$ | $R^2$ without it |
|-----------|-------|---------------|------------------|
| poverty   | 2.718 | 0.1383        | 0.5844           |

|        |       |        |        |
|--------|-------|--------|--------|
| single | 4.799 | 0.3337 | 0.4626 |
| urban  | 0.349 | 0.0026 | 0.6410 |

Each partial  $R^2$  equals  $t^2/(t^2 + 46)$ , and each row satisfies part (a): for poverty,  $1 - 0.6419 = (1 - 0.5844)(1 - 0.1383)$ . Poverty explains 14% of the variation that single parenthood and urbanization leave unexplained, although on its own it explains 36% of the total ( $R^2 = 0.3599$ ). For the pair poverty and urbanization over single parenthood, the  $F$  ratio on 2 and 46 degrees of freedom is 4.502 and the partial  $R^2$  is 0.1637.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")

y = data["murder"].to_numpy()
names = ["poverty", "single", "urban"]
n = len(y)

def fit(cols):
    X = np.column_stack([np.ones(n)] + [data[c].to_numpy() for c in cols])
    return sm.OLS(y, X).fit()

full = fit(names)
df_res = full.df_resid
for j, name in enumerate(names, start=1):
    reduced = fit([c for c in names if c != name])
    partial_r2 = (reduced.ssr - full.ssr) / reduced.ssr
    t = full.tvalues[j]
    print(f"{name:8s} partial R^2 = {partial_r2:.4f}, t = {t:.3f},"
          f" t^2/(t^2 + df) = {t ** 2 / (t ** 2 + df_res):.4f}")
print(f"R^2 = {full.rsquared:.4f}, adjusted R^2 = {full.rsquared_adj:.4f}")
```

[Run this code online](#)

### 9.5.5 What $R^2$ does not measure

A high  $R^2$  is often read as “a good model” and a low one as “a useless model”; neither reading is safe.

**It is not a property of the relationship alone.**  $R^2$  depends on the spread of the regressors, which is a property of the design. In simple regression with fixed  $x$  values, the expectations of the two sums of squares are (Exercise (B1))

$$E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}, \quad E(\text{SST}) = (n-1)\sigma^2 + \beta_1^2 S_{xx},$$

so  $R^2$  is roughly  $\beta_1^2 S_{xx} / (\beta_1^2 S_{xx} + (n-1)\sigma^2)$ . In Figure 9.5.2 the line  $E(y) = 2 + 0.5x$  and the same 50 errors with  $\sigma = 1$  are combined with  $x$  values spread evenly over  $[4, 6]$  and over  $[0, 10]$ . The two designs span the same  $\mathcal{C}(\mathbf{X})$ , so the residuals are identical and the residual mean square is 1.101 in both fits, but  $R^2$  is 0.026 in one case and 0.632 in the other. The approximate values from the formula are 0.081 and 0.689. Restricting the range of a regressor lowers  $R^2$  without changing the relationship.

```
import numpy as np
rng = np.random.default_rng(9)
n, beta0, beta1, sigma = 50, 2.0, 0.5, 1.0
```

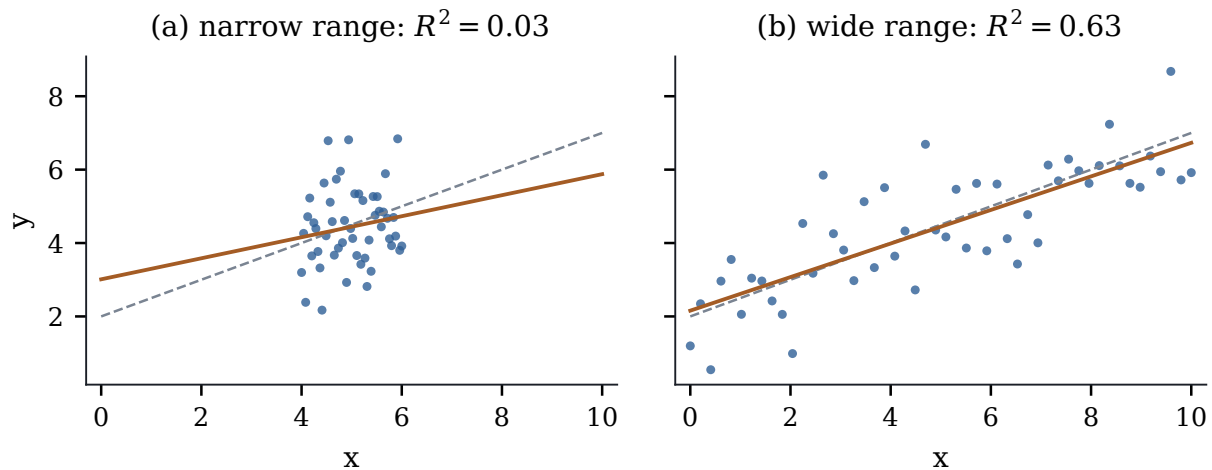


Figure 9.5.2: One line, two designs. The dashed line is the true mean  $2 + 0.5x$ , the solid line the least squares fit. The errors and the residual mean square are the same in both panels.

```

eps = sigma * rng.normal(size=n)                # the same errors in both
designs

def fit(x):
    y = beta0 + beta1 * x + eps
    X = np.column_stack([np.ones(n), x])
    b, *_ = np.linalg.lstsq(X, y, rcond=None)
    sse = np.sum((y - X @ b) ** 2)
    r2 = 1 - sse / np.sum((y - y.mean()) ** 2)
    return y, b, sse / (n - 2), r2

for label, x in [("narrow", np.linspace(4, 6, n)), ("wide", np.linspace(0, 10, n))]:
    y, b, s2, r2 = fit(x)
    Sxx = np.sum((x - x.mean()) ** 2)
    expected = beta1 ** 2 * Sxx / (beta1 ** 2 * Sxx + (n - 1) * sigma ** 2)
    print(f"{label:6s}: slope {b[1]:.3f}, s^2 {s2:.3f}, R^2 {r2:.3f}, design value {
    expected:.3f}")

```

[Run this code online](#)

**It is not a test of the model.** A line fitted to points on a smooth curve can have a high  $R^2$  while missing the curvature; residual plots and lack-of-fit tests (Theorem 14.6.2) reveal it. Conversely, a correct model has a low  $R^2$  whenever the error variance is large, and its coefficients can still be precisely estimated and important.

**It is not comparable across responses.** Models for  $y$  and  $\log y$ , or for individuals and for group averages, have different denominators SST. Averaging replicates inflates  $R^2$  (Exercise (B2)).

**It is not comparable without an intercept** (Proposition 9.1.3, Example (Forcing the murder regression through the origin)).

**It is not a measure of prediction or of cause.** Computed on the fitting data, it overstates predictive accuracy for new data (Chapter 29), and variation “explained” is linear association, not the effect of changing a regressor (Chapter 25).

## 9.5.6 Exercises

### 9.5.6.1 A. Check your understanding

#### Exercise. A1

A regression with an intercept,  $n = 20$  and  $r = 6$  has  $R^2 = 0.40$ . Compute  $\bar{R}^2$  and the overall  $F$  ratio. Is  $\bar{R}^2$  positive? Is it consistent with part (b) of [Proposition 9.5.3](#)?

### 9.5.6.2 B. Practice

#### Exercise. B1

In simple regression with fixed  $x$  values,  $E(\mathbf{Y}) = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , use Theorem 2.3.1 to show that  $E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}$  and  $E(\text{SST}) = (n-1)\sigma^2 + \beta_1^2 S_{xx}$ . Show that the ratio of these expectations tends to 1 as  $S_{xx} \rightarrow \infty$  with  $n$  fixed, and to  $1/(n-1)$  as  $S_{xx} \rightarrow 0$ , and interpret the second limit.

*Solution.*  $\text{SSR} = \mathbf{y}^\top (\mathbf{M} - \mathbf{P}_1) \mathbf{y}$ , where  $\mathbf{M} - \mathbf{P}_1$  projects onto the line spanned by  $\mathbf{x} - \bar{x} \mathbf{1}$  (Example (Intercept inside simple regression)) and has trace 1. With  $\boldsymbol{\mu} = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$ ,  $(\mathbf{M} - \mathbf{P}_1) \boldsymbol{\mu} = \beta_1 (\mathbf{x} - \bar{x} \mathbf{1})$ , whose squared length is  $\beta_1^2 S_{xx}$ . Theorem 2.3.1 gives  $E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}$ . Similarly  $\text{SST} = \mathbf{y}^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{y}$ , with trace  $n-1$  and  $(\mathbf{I} - \mathbf{P}_1) \boldsymbol{\mu} = \beta_1 (\mathbf{x} - \bar{x} \mathbf{1})$ . The ratio  $(\sigma^2 + \beta_1^2 S_{xx}) / ((n-1)\sigma^2 + \beta_1^2 S_{xx})$  tends to 1 as  $S_{xx} \rightarrow \infty$  and to  $1/(n-1)$  as  $S_{xx} \rightarrow 0$ . The second limit is the null value of  $E(R^2)$  for  $r = 2$  ([Theorem 9.5.2](#)): with almost no spread in  $x$ , the slope contributes almost nothing and the regression sum of squares is almost pure noise.

#### Exercise. B2

Suppose each of  $g$  distinct values  $x_1, \dots, x_g$  carries  $m$  observations. Show that the least squares line fitted to all  $gm$  observations is the same as the line fitted to the  $g$  group means, and that its  $R^2$  for the individual data is at most its  $R^2$  for the means, with equality iff every observation equals its group mean.

*Solution.* The model space for the individual data is spanned by  $\mathbf{1}$  and  $\mathbf{x}$ , both constant within groups, so it lies inside the space of group-constant vectors. By Theorem 6.5.1, projecting  $\mathbf{y}$  onto it equals projecting the vector of group means (repeated  $m$  times) onto it, which is the fit to the means with each point weighted equally. Write  $\text{SSR}_g$  and  $\text{SST}_g$  for the sums over the  $g$  means, and  $W$  for the within-group sum of squares. For the individual data  $\text{SSR} = m \text{SSR}_g$  and  $\text{SST} = m \text{SST}_g + W$ . Hence  $R^2 = m \text{SSR}_g / (m \text{SST}_g + W) \leq \text{SSR}_g / \text{SST}_g$ , with equality iff  $W = 0$ .

#### Exercise. B3

Using part (d) of [Theorem 9.5.4](#) and the sequential sums of squares of Example (Order

matters) (order A), express  $1 - R^2$  for the three-regressor model as a product of three factors, and identify each factor as one minus a partial  $R^2$ .

### 9.5.6.3 C. Going deeper

#### Exercise. C1

Under the conditions of [Theorem 9.5.2](#), show that for every  $c \in (0, 1)$  the probability  $\Pr(R^2 > c)$  is a strictly increasing function of the noncentrality  $\gamma$ . *Hint:* part (a) of the theorem and Theorem 4.1.4.

#### Exercise. C2

For two regressors with sample correlations  $r_{y1} = 0.5$ ,  $r_{y2} = 0$  and  $r_{12} = 0.6$ , compute  $R^2$  and compare it with  $r_{y1}^2 + r_{y2}^2$ . Use the formulas of Exercise (B1). Explain geometrically why a regressor that is uncorrelated with the response can raise  $R^2$ .